Energy sources such as solar power, wind, and water are also generally "clean"—unlike fossil fuels, they produce zero or very low greenhouse gas emissions. However, not all renewables are clean. People have burned wood and animal dung for heat and light for hundreds of thousands of years. Trees can be replanted and animals produce more dung, so the practice is sustainable, but burning such fuels also emits carbon dioxide (CO_2), which is one reason why, unlike other forms of renewable energy, they are not classed as "alternative" sources.

Renewable, clean energy will have huge long-term benefits for populations and ecosystems. It reduces pollution, mitigates against global climate change, builds sustainability, and increases the energy security of countries. If it can be provided cheaply enough, it will also pull many people out of poverty. Since 2020, 14 countries have generated more than 95 percent of their electricity from renewable sources every year.

Solar energy

The Sun's power could supply the world's energy needs several times over. The International Energy Agency (IEA) believes that—in the short term—solar power has the greatest potential of all the renewables. The Sun's radiation can be converted directly into electricity via photovoltaic cells (as with solar panels on buildings) or indirectly by using lenses or mirrors to create heat, which can be converted to electricity. This is called concentrated solar power.

Solar panels on a roof can heat domestic water. Sunlight can be employed to desalinate water through an evaporative process, first adopted by 16th-century Arab alchemists and used on an industrial scale in Chile in the late 19th century. In the developing world, solar disinfection is bringing safe drinking water to more than two million people; the process involves using solar heat and ultra-violet light to kill pathogens.